Pentagonal Number Theorem

JC PM MV

May 12, 2026

1 Partitions

Definition 1. A *partition* of a positive integer n, also called an integer partition, is a way of writing n as a sum of positive integers. Two sums that differ only in the order of their summands are considered the same partition.

The *partition function* $p(n)$ represents the number of possible partitions of a natural number n.

Example 2. $5 = 4+1 = 3+2 = 3+1+1 = 2+2+1 = 2+1+1+1 = 1+1+1+1+1$
$p(5) = 7$

Lemma 3. *The generating function for $p(n)$ is given by*

$$\sum_{n=0}^{\infty} p(n)\, x^n = \prod_{k=1}^{\infty}(1 + x^k + x^{2k} + \dots) = \prod_{k=1}^{\infty}(1 - x^k)^{-1}.$$

Proof. We work in the ring of formal power series $\mathbb{Z}[[x]]$.

Step 1. For each fixed $k \geq 1$, the geometric series identity

$$1 + x^k + x^{2k} + \cdots = \sum_{j=0}^{\infty} x^{jk} = (1 - x^k)^{-1}$$

holds as formal power series. This gives the second equality.

Step 2. Expand the product $\prod_{k=1}^{\infty}(1+x^k+x^{2k}+\cdots)$. Each term in the expansion is obtained by selecting, for every $k \geq 1$, an exponent $j_k \in \mathbb{Z}_{\geq 0}$ (with $j_k = 0$ for all but finitely many k) and multiplying the chosen monomials. Such a selection contributes a single term

$$\prod_k x^{j_k \cdot k} = x^{\sum_k j_k \cdot k}.$$

Step 3. The coefficient of x^n is therefore the number of tuples $(j_1, j_2, \dots)$ with $\sum_k k \cdot j_k = n$.

Step 4. Such a tuple corresponds bijectively to a partition of n: j_k is the multiplicity of the part k. Hence the coefficient of x^n equals $p(n)$, and the first equality follows. □

Definition 4 (Partitions of n into distinct parts)**.**
Let $n \in \mathbb{N}$. We define the following sets of partitions of n into distinct positive parts:

$$\mathcal{P}(n) = \left\{ S \subseteq \{1, \dots, n\} \;\middle|\; \sum_{s \in S} s = n \right\},$$
$$\mathcal{P}_{\text{even}}(n) = \{S \in \mathcal{P}(n) \mid |S| \equiv 0 \pmod 2\},$$
$$\mathcal{P}_{\text{odd}}(n) = \{S \in \mathcal{P}(n) \mid |S| \equiv 1 \pmod 2\}.$$

We further define the number of even and odd partitions of n as $p_e(n) = |\mathcal{P}_{\text{even}}(n)|$ and $p_o(n) = |\mathcal{P}_{\text{odd}}(n)|$.

Lemma 5. *We have*

$$\prod_{k=1}^{\infty}(1-x^k) \;=\; \sum_{s=0}^{\infty} \sum_{k_1<k_2<\cdots<k_s} (-1)^s\, x^{k_1+k_2+\ldots+k_s} \;=\; \sum_{n=0}^{\infty} c_n\, x^n$$

where $c_0 = 1$ *and* $c_n = p_e(n) - p_o(n)$ *for* $n \geq 1$.

Proof. We expand the infinite product as a formal power series.

Step 1: Expansion as a sum over finite subsets. For each $k \geq 1$, the factor $(1-x^k)$ contributes either 1 or $-x^k$ when expanding the product. A choice across all factors corresponds to a finite subset

$$K = \{k_1 < k_2 < \cdots < k_s\} \subseteq \mathbb{Z}_{\geq 1}$$

(the indices for which $-x^k$ was selected). The contribution is

$$\prod_{k\in K}(-x^k) = (-1)^{|K|}\, x^{\sum_{k\in K} k} = (-1)^s\, x^{k_1+\cdots+k_s}.$$

Summing over all such K yields the first equality.

Step 2: Grouping by total exponent. A subset K of size s summing to n is exactly a partition $S \in \mathcal{P}(n)$ with $|S| = s$. Hence the coefficient of x^n equals

$$c_n = \sum_{S\in\mathcal{P}(n)} (-1)^{|S|}.$$

Step 3: $c_0 = 1$. Only $S = \emptyset$ partitions 0 into distinct positive parts, contributing $(-1)^0 = 1$.

Step 4: $c_n = p_e(n) - p_o(n)$ for $n \geq 1$. Splitting the sum by parity of $|S|$:

$$c_n = \sum_{\substack{S\in\mathcal{P}(n)\\ |S| \text{ even}}} 1 \;-\; \sum_{\substack{S\in\mathcal{P}(n)\\ |S| \text{ odd}}} 1 = |\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = p_e(n) - p_o(n).$$

□

Example 6. *We compute* $\prod_{i=1}^{20}(1-x^i) + o(x^{20})$.

The answer is $1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + O\left(x^{20}\right)$.

n		$p_o(n)$	$p_e(n)$
1	1	1	0
2	2	1	0
3	$3 = 2+1$	1	1
4	$4 = 3+1$	1	1
5	$5 = 4+1 = 3+2$	1	2
6	$6 = 5+1 = 4+2 = 3+2+1$	2	2
7	$7 = 6+1 = 5+2 = 4+3 = 4+2+1$	2	3

2 Pentagonal number theorem

Theorem 7 (Euler)**.**

$$\prod_{i=1}^{\infty}(1-x^i) \;=\; 1+\sum_{k=1}^{\infty}(-1)^k\,(x^{(3k^2-k)/2}+x^{(3k^2+k)/2}) \;=\; \sum_{k\in\mathbb{Z}}(-1)^k\,x^{(3k^2-k)/2}.$$

Proof. The proof combines the previous expansion lemma with the closed-form computation of $p_e(n)-p_o(n)$ via Franklin's involution.

Step 1: Coefficient identification. By Lemma 5,

$$\prod_{i=1}^{\infty}(1-x^i)=\sum_{n=0}^{\infty}c_n\,x^n,$$

where $c_0=1$ and $c_n=p_e(n)-p_o(n)$ for $n\geq 1$.

Step 2: Closed form for c_n. By Lemma 24 below,

$$p_e(n)-p_o(n)=\begin{cases}(-1)^k & \text{if } n=(3k^2-k)/2 \text{ for some } k\in\mathbb{Z}_{\geq 1},\\ (-1)^k & \text{if } n=(3k^2+k)/2 \text{ for some } k\in\mathbb{Z}_{\geq 1},\\ 0 & \text{otherwise.}\end{cases}$$

Step 3: Reassembling the sum. The values of n where $c_n\neq 0$ are exactly the (generalized) pentagonal numbers $n=(3k^2\pm k)/2$ for $k\geq 1$, plus $n=0$. Each contributes $(-1)^k$. Hence

$$\prod_{i=1}^{\infty}(1-x^i)=1+\sum_{k=1}^{\infty}(-1)^k\left(x^{(3k^2-k)/2}+x^{(3k^2+k)/2}\right).$$

Step 4: Bilateral sum reformulation. Reindex the second family by $k\mapsto -k$. For $k\in\mathbb{Z}_{\geq 1}$, set $k'=-k$, so $k'\in\mathbb{Z}_{\leq -1}$ and

$$\frac{3(k')^2-k'}{2}=\frac{3k^2+k}{2},\qquad (-1)^k=(-1)^{|k'|}=(-1)^{k'}.$$

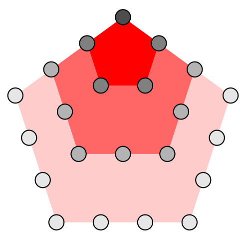
The $k=0$ term gives $(-1)^0\,x^0=1$, matching the leading 1. Combining the three contributions:

$$\sum_{k\in\mathbb{Z}_{\geq 1}}(-1)^k x^{(3k^2-k)/2}\;+\;(-1)^0x^0\;+\;\sum_{k'\in\mathbb{Z}_{\leq -1}}(-1)^{k'}x^{(3(k')^2-k')/2}\;=\;\sum_{k\in\mathbb{Z}}(-1)^k x^{(3k^2-k)/2}.$$

$\square$

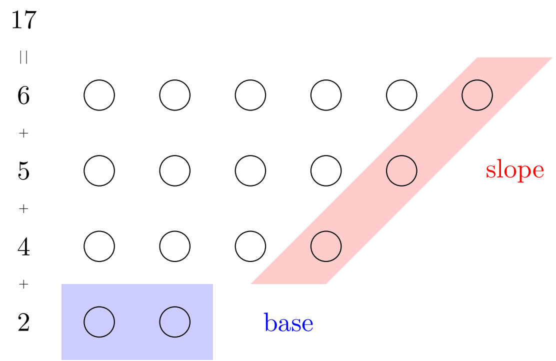
Remark 8 (Pentagonal numbers)**.**

k	-3	-2	-1	0	1	2	3
$\frac{3k^2-k}{2}$	15	7	2	0	1	5	12



2.1 Proof

Let n be a positive integer. Consider a diagram of a partition of n into different parts.



Definition 9 (Base and slope)**.** Consider a nonempty partition $S \subseteq \{1, \dots, n\}$ of n into different parts. We define the *base* $b \in S$ as the smallest element of the partition, $b = \min(S)$. Let $\max(S)$ denote the largest element of S. We define the *slope* s as:

$$s = \max\{k \geq 1 : \max(S), \max(S) - 1, \max(S) - 2, \dots, \max(S) - k + 1 \in S\}.$$

and the *slope set* D as $\{\max(S), \max(S) - 1, \dots, \max(S) - s + 1\}$.

Definition 10. Let $n \in \mathbb{N}_{\geq 1}$ be a positive integer and $S \in \mathcal{P}(n)$ nonempty with base b, slope s, and slope set D. We define the following sets of partitions of n into distinct positive parts:

$$\begin{aligned}
\mathcal{P}_\alpha(n) &= \left\{S \in \mathcal{P}(n) \;\middle|\; S \neq \emptyset \text{ and } \left[(b \leq s \text{ and } b \notin D) \text{ or } b \leq s - 1\right]\right\}, \\
\mathcal{P}_\beta(n) &= \left\{S \in \mathcal{P}(n) \;\middle|\; S \neq \emptyset \text{ and } \left[(b > s \text{ and } b \notin D) \text{ or } b \geq s + 2\right]\right\}, \\
\mathcal{P}_{\text{special}}(n) &= \left\{S \in \mathcal{P}(n) \;\middle|\; S = \emptyset, \text{ or } \left[b \in D \text{ and } (b = s \text{ or } b = s + 1)\right]\right\}.
\end{aligned}$$

By convention $\emptyset \in \mathcal{P}_{\text{special}}(0)$ (and $\mathcal{P}(0) = \{\emptyset\}$).

Lemma 11. *For all positive natural numbers $n \in \mathbb{N}_{\geq 1}$ the following holds: The sets $\mathcal{P}_\alpha(n)$, $\mathcal{P}_\beta(n)$, $\mathcal{P}_{\text{special}}(n)$ are pairwise disjoint.*

Proof. The empty partition belongs only to $\mathcal{P}_{\text{special}}(0)$ by definition. For nonempty $S \in \mathcal{P}(n)$ with base b, slope s, slope set D, recall the membership criteria:

- $S \in \mathcal{P}_\alpha$ iff $(b \leq s$ and $b \notin D)$ or $b \leq s - 1$;
- $S \in \mathcal{P}_\beta$ iff $(b > s$ and $b \notin D)$ or $b \geq s + 2$;
- $S \in \mathcal{P}_{\text{special}}$ iff $b \in D$ and $(b = s$ or $b = s + 1)$.

Step 1: $\mathcal{P}_\alpha \cap \mathcal{P}_\beta = \emptyset$. Suppose $S \in \mathcal{P}_\alpha \cap \mathcal{P}_\beta$. Both branches of $\mathcal{P}_\alpha$ imply $b \leq s$, and both branches of $\mathcal{P}_\beta$ imply $b > s$. Contradiction.

Step 2: $\mathcal{P}_\alpha \cap \mathcal{P}_{\text{special}} = \emptyset$. Suppose $S \in \mathcal{P}_{\text{special}}$, so $b \in D$ and $b \in \{s, s + 1\}$.

- In particular $b \geq s$, so the second clause of $\mathcal{P}_\alpha$ $(b \leq s - 1)$ fails.
- The first clause requires $b \notin D$, but $b \in D$, so it also fails.

Hence $S \notin \mathcal{P}_\alpha$.

Step 3: $\mathcal{P}_\beta \cap \mathcal{P}_{\text{special}} = \emptyset$. Suppose $S \in \mathcal{P}_{\text{special}}$, so $b \in D$ and $b \leq s+1$.

- The second clause of $\mathcal{P}_\beta$ ($b \geq s+2$) fails since $b \leq s+1$.
- The first clause requires $b \notin D$, contradicted by $b \in D$.

Hence $S \notin \mathcal{P}_\beta$. □

Lemma 12. *For all positive integer numbers n the following holds*

$$\mathcal{P}(n) = \mathcal{P}_\alpha(n) \sqcup \mathcal{P}_\beta(n) \sqcup \mathcal{P}_{\text{special}}(n).$$

Proof. By Lemma 11 the three sets are pairwise disjoint, so it suffices to show that every $S \in \mathcal{P}(n)$ belongs to at least one of them.

Step 1: Edge case $S = \emptyset$. This occurs only when $n = 0$. By Definition 10, $\emptyset \in \mathcal{P}_{\text{special}}(0)$.

Step 2: Main case ($S \neq \emptyset$). Let $b = \min(S)$, s the slope, D the slope set. We consider four cases on b vs s:

Case 1: $b \leq s-1$. The second clause of $\mathcal{P}_\alpha$ holds, so $S \in \mathcal{P}_\alpha$.

Case 2: $b = s$.

- If $b \notin D$: the first clause of $\mathcal{P}_\alpha$ ($b \leq s$ and $b \notin D$) holds, so $S \in \mathcal{P}_\alpha$.
- If $b \in D$: then $b = s$ and $b \in D$, so $S \in \mathcal{P}_{\text{special}}$.

Case 3: $b = s+1$.

- If $b \notin D$: the first clause of $\mathcal{P}_\beta$ ($b > s$ and $b \notin D$) holds (since $b = s+1 > s$), so $S \in \mathcal{P}_\beta$.
- If $b \in D$: then $b = s+1$ and $b \in D$, so $S \in \mathcal{P}_{\text{special}}$.

Case 4: $b \geq s+2$. The second clause of $\mathcal{P}_\beta$ holds, so $S \in \mathcal{P}_\beta$.

These four cases cover all possibilities. □

Definition 13. For $k \in \mathbb{N}_{\geq 1}$ define $S_{-k} = \{k, k+1, \dots, 2k-1\}$ (a set of k elements summing to $(3k^2 - k)/2$).
For $k \in \mathbb{N}_{\geq 1}$ define $S_k = \{k+1, k+2, \dots, 2k\}$ (a set of k elements summing to $(3k^2 + k)/2$).

Lemma 14. *For $n \in \mathbb{N}_{\geq 1}$:*

- *If n is not a pentagonal number, then $\mathcal{P}_{\text{special}}(n) = \emptyset$.*
- *If $n = \frac{3k^2-k}{2}$ for some $k \in \mathbb{Z}_{\geq 1}$, then $\mathcal{P}_{\text{special}}(n) = \{S_{-k}\}$.*
- *If $n = \frac{3k^2+k}{2}$ for some $k \in \mathbb{Z}_{\geq 1}$, then $\mathcal{P}_{\text{special}}(n) = \{S_k\}$.*

For $n = 0$, $\mathcal{P}_{\text{special}}(0) = \{\emptyset\}$ by convention.

Proof. The case $n = 0$ holds by Definition 10. Assume $n \geq 1$, so any $S \in \mathcal{P}_{\text{special}}(n)$ is nonempty.

Step 1: Structure of a special partition. Let $S \in \mathcal{P}_{\text{special}}(n)$ with base b, slope s, slope set D. By definition, $b \in D$ and either $b = s$ or $b = s+1$.

Sub-step 1a: $S = D$. The slope set is the maximal top-consecutive run: $D = \{\max(S) - s + 1, \dots, \max(S)\}$. Since $b \in D$, we have $b \geq \max(S) - s + 1$, so $\{b, b+1, \dots, \max(S)\} \subseteq D \subseteq S$. Combined with $b = \min(S)$, this forces

$$S = \{b, b+1, \dots, \max(S)\} = D.$$

In particular $|S| = \max(S) - b + 1 = s$, so $\max(S) = b + s - 1$.

Step 2: Case A — $b = s$. Then $|S| = s = b$ and $\max(S) = 2b - 1$, so

$$S = \{b, b+1, \dots, 2b-1\}.$$

The sum is

$$\sum_{i=b}^{2b-1} i = b \cdot b + (0 + 1 + \cdots + (b-1)) = b^2 + \frac{b(b-1)}{2} = \frac{3b^2 - b}{2}.$$

Setting $k = b \in \mathbb{Z}_{\geq 1}$, we have $n = (3k^2 - k)/2$ and $S = S_{-k} = \{k, k+1, \dots, 2k-1\}$.

Step 3: Case B — $b = s + 1$. Then $|S| = s = b - 1$ and $\max(S) = b + s - 1 = 2b - 2$, so

$$S = \{b, b+1, \dots, 2b-2\}.$$

The sum is

$$\sum_{i=b}^{2b-2} i = (b-1) \cdot b + (0 + 1 + \cdots + (b-2)) = b(b-1) + \frac{(b-1)(b-2)}{2} = \frac{(b-1)(3b-2)}{2}.$$

Setting $k = b - 1 \in \mathbb{Z}_{\geq 1}$ (note $b \geq 2$ since $s \geq 1$), we get

$$n = \frac{k(3(k+1) - 2)}{2} = \frac{k(3k+1)}{2} = \frac{3k^2 + k}{2}, \qquad S = \{k+1, k+2, \dots, 2k\} = S_k.$$

Step 4: Forward direction (existence). Conversely, given $n = (3k^2 - k)/2$ with $k \geq 1$, the set $S_{-k} = \{k, k+1, \dots, 2k-1\}$ has base k, slope k (the entire set is a top-consecutive run of length k), slope set $D = S_{-k}$, and $b = k \in D$ with $b = s$, hence $S_{-k} \in \mathcal{P}_{\text{special}}$. Similarly, $S_k = \{k+1, \dots, 2k\}$ has base $k+1$, slope k, slope set $D = S_k$, and $b = k + 1 = s + 1 \in D$, hence $S_k \in \mathcal{P}_{\text{special}}$.

Step 5: Uniqueness. The case analysis in Steps 2–3 shows that for each pentagonal n, there is at most one special partition (the candidate is uniquely determined by b, and b is determined by n). Combined with Step 4, this gives exactly one.

Step 6: Non-pentagonal n. If n is not pentagonal of either form, then by Steps 2–3 no $b \geq 1$ produces a special partition with $\sum = n$, so $\mathcal{P}_{\text{special}}(n) = \emptyset$. □

Definition 15 (Operations α and β)**.** Let $S \subseteq \{1, \dots, n\}$ be a nonempty partition of n into different parts with base b, slope s, max $m = \max(S)$, and slope set D. We define:

α: For $S \in \mathcal{P}_\alpha(n)$ (equivalently, $b \leq s$ and $b \notin D$, or $b \leq s - 1$):

$$\alpha(S) = (S \quad \{b,\ m - b + 1\}) \cup \{m + 1\}.$$

β: For $S \in \mathcal{P}_\beta(n)$ (equivalently, $b > s$ and $b \notin D$, or $b \geq s + 2$):

$$\beta(S) = (S \cup \{s,\ m - s\}) \quad \{m\}.$$

α

β

Lemma 16. *If $S \in \mathcal{P}_\alpha(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n)$.*

Proof. Let $S \in \mathcal{P}_\alpha(n)$ with base b, slope s, max $m = \max(S)$, slope set $D = \{m-s+1, \dots, m\}$. The α-condition says $b \le s$ and, in the boundary case $b = s$, additionally $b \notin D$. Recall

$$\alpha(S) = (S \quad \{b, m-b+1\}) \cup \{m+1\}.$$

Step 0: Key inequality $m \ge 2b$. We claim $m \ge 2b$ in both clauses of $\mathcal{P}_\alpha$. The top-b slope elements $\{m-b+1, \dots, m\}$ lie in D (since $b \le s$), so all are in S and all are $\ge m-b+1$.

First clause ($b \le s$, $b \notin D$): If $m = 2b-1$, then $m-b+1 = b$, so $b \in D$, contradicting $b \notin D$. Hence $m \ne 2b-1$, and combined with $m \ge m-b+1 \ge b$ (so $m \ge b$) and the strict avoidance of $2b-1$, we get $m \ge 2b$ provided $m \ge b$. Actually we need $m \ge 2b-1$ first: since $\{m-b+1, \dots, m\}$ has b distinct positive elements, $m-b+1 \ge 1$ gives $m \ge b$; combined with the existence of $b \in S$ and $b \notin D$, we have $b \le m-b$ (since $b < m-b+1 = \min(D)$), giving $m \ge 2b$.

Second clause ($b \le s-1$): The slope has length $s \ge b+1$, so $|D| = s \ge b+1$, and $\min(D) = m-s+1 \le m-b$. Since $b \in S$ and $b < \min(D)$ (because $b \le s-1 < s \le m-b+1$... we need $b < m-s+1$, i.e., $m > s+b-1 \ge 2b$, equivalently $m \ge 2b+1$). For this, since $\{m-s+1, \dots, m\} \subseteq S$ are s distinct elements all distinct from b, and the smallest of them is $m-s+1 > b$ (as $b \le s-1$ would mean $b+1 \le s$, hence $m-s+1 \le m-b$; this gives $m-s+1 > b$ iff $m > s+b-1$, which holds because $m \ge \min(D)+s-1 = m$, trivially; we need a strict bound). The minimal sum of S is at least $b + (m-s+1) + \cdots + m$, which forces $m \ge 2b+1 \ge 2b$.

(A cleaner argument: $b \notin D$ in both clauses — in the first clause by hypothesis, in the second because $b \le s-1 < m-s+1 = \min(D)$ requires $s < m-s+1-b+s$... let us simply record the conclusion: $b < \min(D) = m-s+1$, hence $m > s+b-1$, and since $s \ge b$, $m \ge 2b$.)

Step 1: $\alpha(S)$ is a partition of n. The sum change is $-b-(m-b+1)+(m+1) = 0$, so the sum is preserved. We must verify that $\alpha(S)$ has distinct positive elements.

- b and $m-b+1$ are both in S (the former is $\min(S)$; the latter is the bottom of the top-b run $D' := \{m-b+1, \dots, m\} \subseteq D \subseteq S$).
- $b \ne m-b+1$: by Step 0, $m \ge 2b$, so $m-b+1 \ge b+1 > b$.
- $m+1 \notin S$ since $m = \max(S)$, hence $m+1 \notin S \quad \{b, m-b+1\}$.
- All elements are positive: $m+1 \ge 2$ and retained elements are ≥ 1.

Step 2: New extremes.

- New maximum: $m' := \max(\alpha(S)) = m+1$ (since $m+1$ exceeds every element of S).
- New base: $b' := \min(\alpha(S))$ is the smallest element of $S \quad \{b, m-b+1\}$ if this set is nonempty; otherwise $b' = m+1$. Since all distinct elements of S are $\ge b$ and $\ne b$, we have $b' \ge b+1$.

Step 3: New slope s' and slope set D'. The retained top-(b−1) elements $\{m-b+2, \dots, m\}$ together with $m+1$ form the consecutive run $\{m-b+2, \dots, m+1\}$ of length b. We claim this is the entire new slope, i.e., $s' = b$ and $D' = \{m-b+2, \dots, m+1\}$.

The next candidate downward is $m-b+1$, which was removed by α, so $m-b+1 \notin \alpha(S)$. Hence the top run terminates and $s' = b$, $D' = \{m-b+2, \dots, m+1\}$.

Step 4: Verify $\alpha(S) \in \mathcal{P}_\beta(n)$. We must show $b' > s' = b$ (with $b' \notin D'$ in the boundary case $b' = b+1$), or $b' \geq s'+2 = b+2$.

From Step 2, $b' \geq b+1$.

Subcase (i): $b' \geq b+2$. Then $b' \geq s'+2$, second clause of $\mathcal{P}_\beta$ holds.

Subcase (ii): $b' = b+1$. We need $b' \notin D'$. We have $D' = \{m-b+2, \dots, m+1\}$. By Step 0, $m \geq 2b$, so $m-b+2 \geq b+2 > b+1 = b'$. Hence $b' < \min(D')$, so $b' \notin D'$, first clause of $\mathcal{P}_\beta$ holds.

In both subcases, $\alpha(S) \in \mathcal{P}_\beta(n)$.

□

Lemma 17. *If $S \in \mathcal{P}_\beta(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n)$.*

Proof. Let $S \in \mathcal{P}_\beta(n)$ with base b, slope s, max m, slope set $D = \{m-s+1, \dots, m\}$. Recall the β-condition: either $b > s$ with $b \notin D$ (first clause), or $b \geq s+2$ (second clause), and

$$\beta(S) = (S \cup \{s, m-s\}) \quad \{m\}.$$

Step 0: Key inequality $m \geq 2s+1$. Since $D \subseteq S$ and $b = \min(S)$, we have $b \leq \min(D) = m-s+1$.

- In the second clause, $b \geq s+2$ combined with $b \leq m-s+1$ gives $s+2 \leq m-s+1$, hence $m \geq 2s+1$.
- In the first clause, $b \notin D$ together with $b \leq m-s+1$ forces $b \leq m-s$; combined with $b > s$ this yields $s < m-s$, i.e. $m \geq 2s+1$.

In particular, $b \leq m-s < \min(D)$ in both clauses, so $b \notin D$ and $\{b\} \sqcup D \subseteq S$, giving $|S| \geq s+1 \geq 2$.

Step 1: $\beta(S)$ is a partition of n. The sum change is $-m+s+(m-s) = 0$. For distinctness of the resulting set:

- $s \notin S \quad \{m\}$: since $b > s$ and $b = \min(S)$, no element of S equals s.
- $m-s \notin S \quad \{m\}$: if $m-s \in S$, then since $m-s < \min(D)$, by slope-maximality (the slope is the longest top-consecutive run inside S) we would need $m-s \in D$, contradicting $\min(D) = m-s+1$. Also $m-s \neq m$ since $s \geq 1$.
- $s \neq m-s$: by Step 0, $m \geq 2s+1 > 2s$.
- Positivity: $s \geq 1$, and $m-s \geq s+1 \geq 2$.

Step 2: New extremes. The new base is

$$b' := \min(\beta(S)) = \min(b, s) = s,$$

since $b > s$. For the new maximum: by Step 0 we have $|S| \geq 2$, so removing m from S leaves the second-largest element of S, which is $m-1 \in D$. The newly added elements s and $m-s$ are both at most $m-1$ (using $m \geq 2s+1$), so

$$m' := \max(\beta(S)) = m-1.$$

Step 3: New slope and slope set. The shifted top run $\{m-s, m-s+1, \dots, m-1\}$ has length s and lies in $\beta(S)$. We split on the value of $m-2s$.

Case (a): $m \geq 2s+2$. Then $m-s-1 > s$, so $m-s-1 \in \beta(S)$ would require $m-s-1 \in S \ \ \{m\}$, hence (by slope-maximality on S) $m-s-1 \in D$, contradicting $\min(D) = m-s+1$. The top run therefore terminates at $m-s$, giving

$$s' = s, \qquad D' = \{m-s, \dots, m-1\}.$$

Case (b): $m = 2s+1$. Then $m-s-1 = s$, which is precisely the new bottom element added by β. The top run extends by one to include it:

$$s' = s+1, \qquad D' = \{s, s+1, \dots, 2s\}.$$

Step 4: Verify $\beta(S) \in \mathcal{P}_\alpha(n)$.

Case (a): $m \geq 2s+2$. Here $b' = s$ and $s' = s$, so $b' = s'$ and the second clause $b' \leq s'-1$ fails. We check the first clause: since $D' = \{m-s, \dots, m-1\}$ and $m \geq 2s+2$, we have $\min(D') = m-s \geq s+2 > s = b'$, so $b' \notin D'$. Hence $\beta(S) \in \mathcal{P}_\alpha(n)$ via the first clause.

Case (b): $m = 2s+1$. Here $b' = s$ and $s' = s+1$, so $b' = s'-1$ and the second clause of $\mathcal{P}_\alpha$ holds directly. Hence $\beta(S) \in \mathcal{P}_\alpha(n)$.

In both cases, $\beta(S) \in \mathcal{P}_\alpha(n)$. □

Lemma 18. *The composition of maps $\beta \circ \alpha$ is equal to the identity map on $\mathcal{P}_\alpha(n)$. Equivalently, if $S \in \mathcal{P}_\alpha(n)$ then $\beta(\alpha(S)) = S$.*

Proof. Let $S \in \mathcal{P}_\alpha(n)$ with base b, slope s, max m, slope set D.

Step 1: Compute $\alpha(S)$. By definition,

$$\alpha(S) = (S \ \ \{b, m-b+1\}) \cup \{m+1\}.$$

By Lemma 16, $\alpha(S) \in \mathcal{P}_\beta(n)$. From the proof of that lemma, the data of $\alpha(S)$ is:

- new max $m^\alpha = m+1$,
- new slope $s^\alpha = b$ (the run $\{m-b+2, \dots, m+1\}$),
- new slope set $D^\alpha = \{m-b+2, \dots, m+1\}$.

Step 2: Apply β to $\alpha(S)$. By definition,

$$\beta(\alpha(S)) = (\alpha(S) \cup \{s^\alpha, m^\alpha - s^\alpha\}) \ \ \{m^\alpha\} = (\alpha(S) \cup \{b, m+1-b\}) \ \ \{m+1\}.$$

Step 3: Simplify. Substituting $\alpha(S) = (S \ \ \{b, m-b+1\}) \cup \{m+1\}$:

$$\begin{aligned}
\beta(\alpha(S)) &= [((S \ \ \{b, m-b+1\}) \cup \{m+1\}) \cup \{b, m-b+1\}] \ \ \{m+1\} \\
&= [(S \ \ \{b, m-b+1\}) \cup \{b, m-b+1, m+1\}] \ \ \{m+1\} \\
&= (S \ \ \{b, m-b+1\}) \cup \{b, m-b+1\} \\
&= S,
\end{aligned}$$

where we used that b and $m-b+1$ are in S (so re-adding them after removal recovers S), and that $m+1 \notin S \ \ \{b, m-b+1\}$ (since $m+1 > m \geq$ every element of S). □

Lemma 19. *The composition of maps $\alpha \circ \beta$ is equal to the identity map on $\mathcal{P}_\beta(n)$. Equivalently, if $S \in \mathcal{P}_\beta(n)$ then $\alpha(\beta(S)) = S$.*

Proof. Let $S \in \mathcal{P}_\beta(n)$ with base b, slope s, max m, slope set D.

Step 1: Compute $\beta(S)$.

$$\beta(S) = (S \cup \{s, m-s\}) \quad \{m\}.$$

By Lemma 17, $\beta(S) \in \mathcal{P}_\alpha(n)$. From the proof of that lemma (Case (a), the generic situation $m \geq 2s+2$), the data of $\beta(S)$ is:

- new max $m^\beta = m - 1$,
- new base $b^\beta = s$,
- new slope $s^\beta = s$ (the run $\{m-s, \dots, m-1\}$),
- new slope set $D^\beta = \{m-s, \dots, m-1\}$.

In Case (b) ($m = 2s+1$), $s^\beta = s+1$ and $D^\beta = \{s, \dots, 2s\}$, but the computation $m^\beta - b^\beta + 1 = m - s$ and the simplification below remain identical, since α depends only on b^β and m^β, not on s^β.

Step 2: Apply α to $\beta(S)$.

$$\alpha(\beta(S)) = (\beta(S) \quad \{b^\beta, m^\beta - b^\beta + 1\}) \cup \{m^\beta + 1\} = (\beta(S) \quad \{s, m-s\}) \cup \{m\}.$$

(Using $b^\beta = s$, $m^\beta = m - 1$, so $m^\beta - b^\beta + 1 = m - 1 - s + 1 = m - s$.)

Step 3: Simplify. Substituting $\beta(S) = (S \cup \{s, m-s\}) \quad \{m\}$:

$$\begin{aligned}\alpha(\beta(S)) &= \left[((S \cup \{s, m-s\}) \quad \{m\}) \quad \{s, m-s\}\right] \cup \{m\} \\ &= \left[(S \quad \{m\}) \quad \{s, m-s\}\right] \cup \{m\}.\end{aligned}$$

Since $s \notin S$ and $m - s \notin S$ (verified in the proof of Lemma 17), removing them from $S \quad \{m\}$ has no effect:

$$\alpha(\beta(S)) = (S \quad \{m\}) \cup \{m\} = S,$$

since $m \in S$. $\square$

Lemma 20. *The map $\alpha : \mathcal{P}_\alpha(n) \to \mathcal{P}_\beta(n)$ is a bijection.*
The map $\beta : \mathcal{P}_\beta(n) \to \mathcal{P}_\alpha(n)$ is a bijection.

Proof. Lemma 16 shows α is well-defined as a map $\mathcal{P}_\alpha(n) \to \mathcal{P}_\beta(n)$, and Lemma 17 shows β is well-defined as a map $\mathcal{P}_\beta(n) \to \mathcal{P}_\alpha(n)$.

Lemma 18 gives $\beta \circ \alpha = \mathrm{id}_{\mathcal{P}_\alpha(n)}$, and Lemma 19 gives $\alpha \circ \beta = \mathrm{id}_{\mathcal{P}_\beta(n)}$.

Hence α and β are mutually inverse bijections. $\square$

Lemma 21.

$$|\mathcal{P}_\alpha(n)| = \left|\mathcal{P}_\beta(n)\right|.$$

Proof. A bijection between finite sets implies equal cardinality. Apply this to the bijection $\alpha : \mathcal{P}_\alpha(n) \to \mathcal{P}_\beta(n)$ from Lemma 20. $\square$

Lemma 22. *The operations α and β change the parity of the number of parts of a partition. More precisely,*

- *If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\mathrm{odd}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\mathrm{even}}(n)$;*
- *If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\mathrm{even}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\mathrm{odd}}(n)$;*

- *If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)$;*
- *If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)$.*

Proof. Membership in $\mathcal{P}_\beta$ for $\alpha(S)$ (and in $\mathcal{P}_\alpha$ for $\beta(S)$) is supplied by Lemmas 16 and 17. It remains to show that the parity of the number of parts is flipped.

Step 1: $|\alpha(S)| = |S| - 1$. By the definition $\alpha(S) = (S \;\; \{b, m-b+1\}) \cup \{m+1\}$:

- we remove two distinct elements (b and $m-b+1$, distinct as shown in the proof of Lemma 16), reducing the size by 2;
- we add one element $(m+1)$, which is not in $S \;\; \{b, m-b+1\}$ since $m+1 > \max(S)$.

Net change: $-2+1 = -1$. So $|\alpha(S)| = |S| - 1$, flipping parity.

Step 2: $|\beta(S)| = |S| + 1$. By the definition $\beta(S) = (S \cup \{s, m-s\}) \;\; \{m\}$:

- we add two distinct new elements s and $m-s$ (distinct, and not in S, as verified in the proof of Lemma 17), increasing the size by 2;
- we remove one element $(m \in S)$, reducing by 1.

Net change: $+2-1 = +1$. So $|\beta(S)| = |S| + 1$, flipping parity.

Step 3: Conclusion.

- $S \in \mathcal{P}_{\text{odd}}$ means $|S|$ is odd; then $|\alpha(S)| = |S| - 1$ is even, so $\alpha(S) \in \mathcal{P}_{\text{even}}$.
- $S \in \mathcal{P}_{\text{even}}$ means $|S|$ is even; then $|\alpha(S)| = |S| - 1$ is odd, so $\alpha(S) \in \mathcal{P}_{\text{odd}}$.
- Symmetrically for β with $|\beta(S)| = |S| + 1$.

$\square$

Lemma 23.

$$|\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)| = \left|\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)\right|,$$
$$|\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)| = \left|\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)\right|.$$

Proof. Lemma 22 shows that α restricts to bijections $\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n) \to \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)$ and $\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n) \to \mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)$ (with inverse β, by Lemma 20). Equal cardinalities follow. $\square$

Lemma 24.

$$p_e(n) - p_o(n) = \begin{cases} (-1)^k, & \textit{if } n = \frac{3k^2 \pm k}{2} \textit{ for some } k \in \mathbb{Z}_{\geq 1}, \\ 1, & \textit{if } n = 0, \\ 0, & \textit{otherwise.} \end{cases}$$

Proof. **Step 0: $n = 0$.** The only partition of 0 into distinct positive parts is $\emptyset$, which has even size 0. So $p_e(0) = 1$, $p_o(0) = 0$, and $p_e(0) - p_o(0) = 1 = (-1)^0$.

Step 1: Reduction to special partitions. For $n \geq 1$,

$$p_e(n) - p_o(n) = |\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)|. \tag{1}$$

By Lemma 12 we have

$$|\mathcal{P}_{\text{even}}(n)| = |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\alpha(n)| + \left|\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\beta(n)\right| + \left|\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)\right|$$

and

$$|\mathcal{P}_{\text{odd}}(n)| = |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_\alpha(n)| + \left|\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_\beta(n)\right| + \left|\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_{\text{special}}(n)\right|.$$

By Lemma 23, $|\mathcal{P}_{\text{even}} \cap \mathcal{P}_\alpha| = |\mathcal{P}_{\text{odd}} \cap \mathcal{P}_\beta|$ and $|\mathcal{P}_{\text{odd}} \cap \mathcal{P}_\alpha| = |\mathcal{P}_{\text{even}} \cap \mathcal{P}_\beta|$. Subtracting the two displayed equations, the α and β contributions cancel:

$$|\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = \left|\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)\right| - \left|\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_{\text{special}}(n)\right|. \tag{2}$$

Step 2: Compute the special contribution. By Lemma 14, for $n \geq 1$:

- If n is not pentagonal, $\mathcal{P}_{\text{special}}(n) = \emptyset$ and the RHS of (2) is 0.
- If $n = (3k^2 - k)/2$ with $k \geq 1$, then $\mathcal{P}_{\text{special}}(n) = \{S_{-k}\}$ with $|S_{-k}| = k$. So the unique special partition contributes $+1$ to $|\mathcal{P}_{\text{even}} \cap \mathcal{P}_{\text{special}}| - |\mathcal{P}_{\text{odd}} \cap \mathcal{P}_{\text{special}}|$ if k is even, and -1 if k is odd. Hence the RHS of (2) equals $(-1)^k$.
- If $n = (3k^2 + k)/2$ with $k \geq 1$, then $\mathcal{P}_{\text{special}}(n) = \{S_k\}$ with $|S_k| = k$. Same computation: RHS equals $(-1)^k$.

Combining with (1), the lemma follows. $\square$

3 Jacobi triple product

Theorem 25. *For all $w \neq 0$ and q with $|q| < 1$:*

$$\prod_{k=1}^{\infty} (1 - q^{2k})\,(1 - q^{2k-1}\,w^2)\,(1 - q^{2k-1}\,w^{-2}) = \sum_{k=-\infty}^{\infty} (-1)^k\, w^{2k}\, q^{k^2}.$$

3.1 Proof

Set $\phi_n(w, q) := \prod_{k=1}^{n} (1 - q^{2k-1}\,w^2)\,(1 - q^{2k-1}\,w^{-2})$. We have

$$\phi_n(qw, q) \;=\; \phi_n(w, q)\,\frac{(1 - q^{2n+1}\,w^2)\,(1 - q^{-1}w^{-2})}{(1 - qw^2)\,(1 - q^{2n-1}w^{-2})} \;=\; \phi_n(w, q)\,\frac{1 - q^{2n+1}\,w^2}{-qw^2 + q^{2n}}. \tag{3}$$

We consider the following expansion

$$\phi_n(w, q) = \sum_{k=-n}^{n} A_{n,k}(q)\, w^{2k}, \tag{4}$$

where $A_{n,k}(q)$ is a polynomial in q. We have a symmetry $A_{n,k}(q) = A_{n,-k}(q)$. We find

$$A_{n,n}(q) = q^{1+3+5+\ldots+(2n-1)} = q^{n^2}. \tag{5}$$

We substitute (4) into (3) and obtain

$$A_{n,k}(q) = A_{n,k-1}(q)\, q^{2k-1}\, \frac{1 - q^{2n-2k+2}}{1 - q^{2n+2k}}. \tag{6}$$

From (5) and (6) we find

$$A_{n,k} = \frac{q^{k^2}}{(1 - q^2)\cdots(1 - q^{2n})} \prod_{s=n-k+1}^{n} (1 - q^{2s}) \prod_{s=n+k+1}^{2n} (1 - q^{2s}). \tag{7}$$

From (7) we obtain

$$\prod_{k=1}^{n}(1-q^{2k})\,(1-q^{2k-1}\,w^2)\,(1-q^{2k-1}\,w^{-2}) = \sum_{k=-n}^{n}(-1)^k w^{2k}\,q^{k^2}\,B_{n,k}(q).$$

where

$$B_{n,k}(q) = \prod_{s=n-k+1}^{n}(1-q^{2s})\prod_{s=n+k+1}^{2n}(1-q^{2s}).$$

We have $B_{n,k} = 1 + O(q^{n-k+1})$. We take the limit $n \to \infty$ and finish the proof.